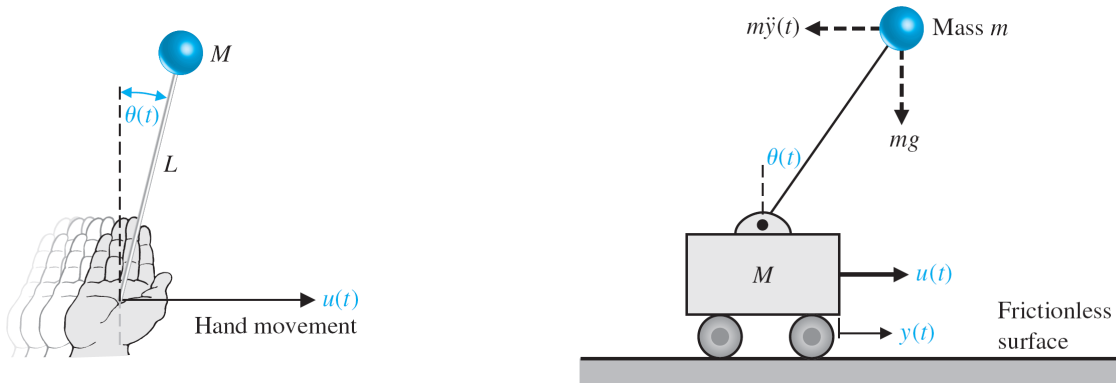


Tutorial - 5

1. The problem of balancing a broomstick on a person's hand is illustrated in the below figure. The only equilibrium condition is $\theta(t) = 0$ and $\dot{\theta}(t) = 0$. The problem of balancing a broomstick on one's hand is not unlike the problem of controlling the attitude of a missile during the initial stages of launch. This problem is the classic and intriguing problem of the inverted pendulum mounted on a cart, as shown in the right figure below. The cart must be moved so that mass m is always in an upright position. The state variables must be expressed in terms of the angular rotation $\theta(t)$ and the position of the cart $y(t)$. The differential equations describing the motion of the system can be obtained by writing the sum of the forces in the horizontal direction and the sum of the moments about the pivot point. Obtain the mathematical model for the given system and linearize it around the equilibrium point.



2. The characteristic polynomial of a system is $q(s) = s^4 + 2s^3 + 3s^2 + 4s + 5$. Comment on the stability of the system using Routh-Hurwitz stability criterion.
3. The open-loop transfer function of a unity negative feedback system is $G(s) = \frac{K}{s(s^2 + s + 2)(s + 3)}$. Find the range of K for which the closed-loop system is stable.
4. The open-loop transfer function of a control system is given as $G(s) = \frac{4}{s^5 + s^4 + 3s^3 + 3s^2 + 2s + 1}$. Determine the number of closed-loop poles in the open left half and in the open right half of the complex plane.
5. Consider the transfer function $G(s) = \frac{2s^2 + 20s + 17}{s^3 + 7s^2 + 25s + 39}$. Determine the number of poles of $G(s)$ having their real parts within the interval $(-2.5, -1)$.
6. In the past, Type-1 diabetes patients had to inject themselves with insulin three to four times a day. New delayed-action insulin analogues such as insulin Glargine require a single daily dose. For a specific patient, state-space model for the concentration-time evolution of plasma for insulin

Glargine is given by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -0.435 & 0.209 & 0.02 \\ 0.268 & -0.394 & 0 \\ 0.227 & 0 & -0.02 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u.$$

The state variables are:

x_1 : insulin amount in plasma compartment

x_2 : insulin amount in liver compartment

x_3 : insulin amount in interstitial (in body tissue) compartment.

The system's input is u = external insulin flow and its output y is plasma insulin concentration such that

$$y = 0.0003x_1.$$

Use the final value theorem to calculate the steady-state error for unit impulse, unit step and unit ramp inputs.

7. An electric ventricular assist device (EVAD) has been designed to help patients with diminished but still functional heart-pumping action to work in parallel with the natural heart. The device consists of a brushless dc electric motor that actuates on a pusher plate. The plate movements help the ejection of blood in systole and sac filling in diastole. System dynamics during systolic mode have been found to be:

$$\begin{bmatrix} \dot{x} \\ \dot{v} \\ \dot{P}_{ao} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -68.3 & -7.2 \\ 0 & 3.2 & -0.7 \end{bmatrix} \begin{bmatrix} x \\ v \\ P_{ao} \end{bmatrix} + \begin{bmatrix} 0 \\ 425.4 \\ 0 \end{bmatrix} e_m.$$

The state variables in this model are x , the pusher plate position; v , the pusher plate velocity; and P_{ao} , the aortic blood pressure. The input to the system is e_m , the motor voltage.

- From the eigenvalues obtained, comment on the stability of the system.
- If the blood pressure is the measured output, Determine the steady-state error for a unit step input.

8. For the system shown in the following figure, find the *system type* for the loop transfer function, the appropriate error constant K_p , and the steady-state error for a unit step input. (Assume input and output signals are of same unit.)

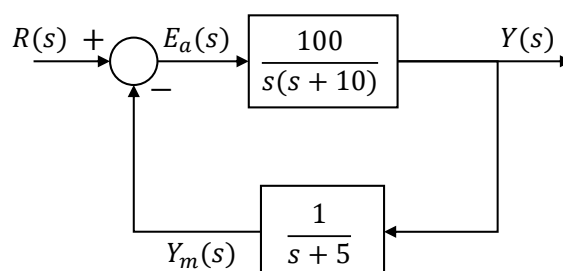


Figure 1

9. Find the total steady-state error due to a unit step input and a unit step disturbance in the following system.

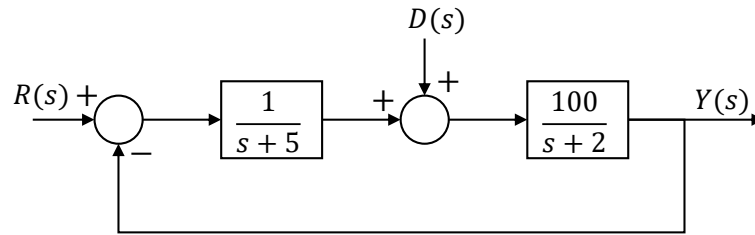


Figure 2

10. Given the system shown below, find the sensitivity of the steady-state error to parameter ' a '; assume a unit step input.

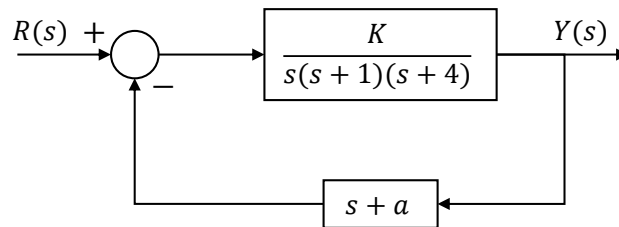


Figure 3